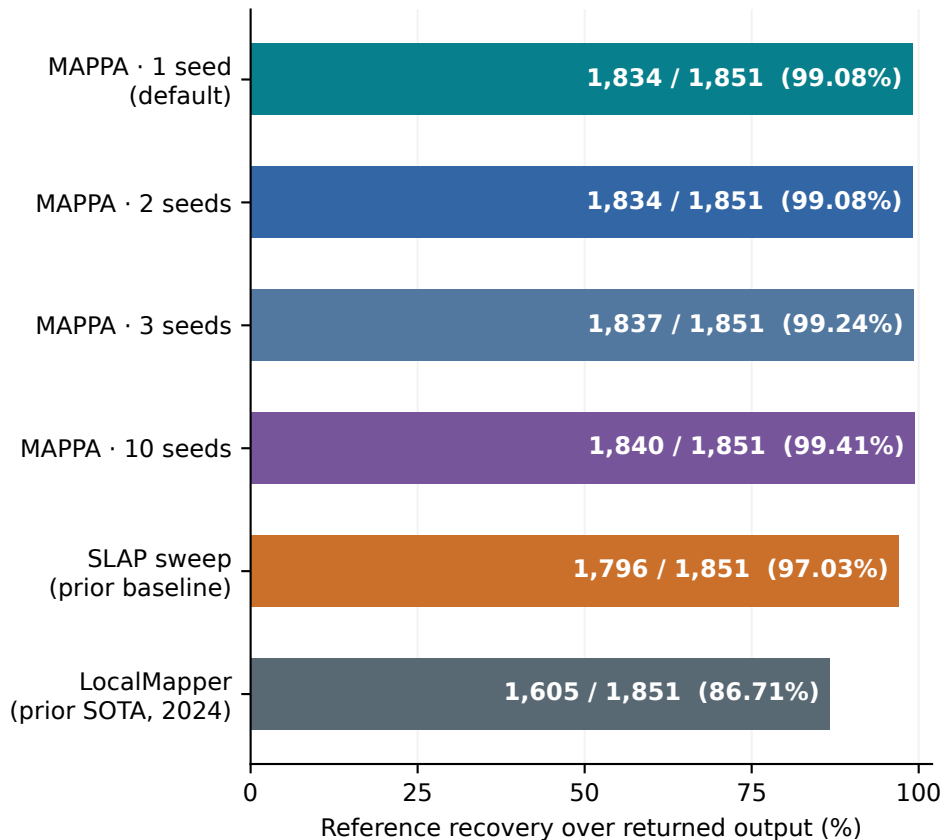


a MAPPA search-stage recovery**b Search cost by size orientation**